

DS UNIT-4

1. What are the key components of supervised learning?

A. Key Components of Supervised Learning

1. Labeled Dataset

- Contains input features (independent variables) and corresponding output labels (dependent variable).
- Example: In predicting house prices, features might be square footage and number of rooms, while the label is the actual price.

2. Training Process

- The model learns patterns by minimizing the difference between predicted outputs and actual labels.
- Data is usually split into training and testing sets to evaluate generalization.

3. Learning Algorithm

- Algorithms define how the model maps inputs to outputs.
- Common examples: Linear Regression, Logistic Regression, Decision Trees, Support Vector Machines, Neural Networks.

4. Model Evaluation

- Performance is measured using metrics appropriate to the task.
- For classification: accuracy, precision, recall, F1-score.
- For regression: mean squared error, mean absolute error, R^2 .

5. Prediction/Deployment

- Once trained and validated, the model is used to predict outcomes on new, unseen data.
- Example: Predicting whether a new email is spam or not.

2. Why is generalization important in machine learning?

A.

1. **Definition** Generalization is the ability of a model to perform well on new, unseen data, not just the training set.
2. **Real-World Usefulness** Models are deployed on future or unknown data. Without generalization, they fail outside the training environment.
3. **Avoiding Overfitting** Overfitting happens when a model memorizes training data instead of learning patterns. Generalization prevents this.
4. **Robustness** A generalized model adapts to variations in data distribution, making it reliable across different scenarios.
5. **Decision-Making Reliability** In critical fields like healthcare, finance, or autonomous systems, poor generalization can lead to costly or dangerous errors.

3. Define Random Forest in machine learning?

A.

1. Core Idea

- A Random Forest consists of **many decision trees** trained on different random subsets of the data.
- Each tree makes its own prediction, and the forest combines them:
 - **Classification:** Uses majority voting among trees.
 - **Regression:** Uses averaging of tree outputs.

2. Randomness in Training

- **Bootstrap Sampling:** Each tree is trained on a random sample of the dataset (with replacement).
- **Feature Randomness:** At each split in a tree, only a random subset of features is considered.
- These two sources of randomness make the trees diverse, which improves generalization.

3. Advantages

- **High Accuracy:** By aggregating many trees, Random Forest reduces variance compared to a single decision tree.
- **Robustness:** Resistant to overfitting, especially with large datasets.
- **Versatility:** Works well for both classification (e.g., spam detection) and regression (e.g., predicting house prices).
- **Feature Importance:** Provides insights into which features are most influential in predictions.

4. Limitations

- **Complexity:** More computationally intensive than a single decision tree.
- **Interpretability:** Harder to interpret compared to a single tree, since predictions come from many trees.

4. Mention one application of logistic regression in real life?

A. Here are some **real-life applications of Logistic Regression**

1. **Medical Diagnosis** – Predicting whether a patient has a particular disease (e.g., diabetes, cancer) based on symptoms and test results.
2. **Banking and Finance** – Determining whether a customer is likely to default on a loan or not (credit risk analysis).
3. **Spam Detection** – Classifying emails as “spam” or “not spam.”
4. **Marketing** – Predicting whether a customer will buy a product (yes/no) based on demographics and purchase history.
5. **Insurance Industry** – Estimating the probability of an accident or claim occurrence.
6. **Human Resources** – Predicting whether an employee will leave a company (employee attrition).
7. **Legal/Criminal Justice** – Predicting the likelihood of a criminal reoffending (recidivism prediction).

5. Discuss why splitting the dataset into training, validation, and test sets is

necessary for supervised learning?

A. 1. Training Set

- Used to **fit the model**—the algorithm learns patterns from this data.
- Contains both input features and corresponding labels.
- The model adjusts its internal parameters (e.g., weights in linear regression) based on this data.

2. Validation Set

- Used to **tune hyperparameters** and make decisions about model architecture.
- Helps monitor **overfitting**: if performance on training is high but poor on validation, the model is memorizing rather than generalizing.
- Example: Choosing the number of trees in a random forest or the learning rate in gradient descent.

3. Test Set

- Used only at the **final evaluation stage**.
- Provides an **unbiased estimate** of model performance on unseen data.
- Ensures that the model's reported accuracy is not inflated by repeated tuning.

6. Discuss two major types of regression models used in data science?

A. 1. Linear Regression

- **Purpose:** Predicts a *continuous* numerical outcome.
- **Model Form:** $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon$
- **Assumptions:**
 - Linearity between predictors and response
 - Homoscedasticity (constant variance of errors)
 - Independence of errors
 - Normal distribution of errors
- **Use Cases:**
 - Predicting house prices based on features like area, location, and number of rooms
 - Forecasting sales or revenue
- **Variants:**
 - *Multiple Linear Regression*: More than one predictor
 - *Ridge and Lasso Regression*: Regularized versions to handle multicollinearity and overfitting

2. Logistic Regression

- **Purpose:** Predicts a *categorical* outcome, typically binary (e.g., 0 or 1).
- **Model Form:** Uses the logistic (sigmoid) function to map predicted values to probabilities:

$$P(y = 1 | x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}}$$

- **Assumptions:**
 - Linearity between independent variables and the log-odds of the dependent variable
 - No multicollinearity

- Large sample size for stable estimates
- **Use Cases:**
 - Spam detection (spam vs. not spam)
 - Predicting customer churn (yes/no)
 - Medical diagnosis (disease present or not)

7. What is logistic regression? Explain its working with the logistic (sigmoid) function and graph?

A. **Logistic regression** is a supervised learning algorithm used for **binary classification** problems. It predicts the probability that a given input belongs to one of two classes (e.g., spam vs. not spam, disease vs. no disease).

1. Core Idea

Unlike linear regression which predicts continuous values, logistic regression predicts a **probability** between 0 and 1. It uses the **logistic (sigmoid) function** to map any real-valued input to this probability range.

2. Mathematical Model

Given input features $x_1, x_2, \dots, x_n$, logistic regression computes:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Then applies the **sigmoid function**:

$$P(y = 1 | x) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

This output P is interpreted as the probability that the input belongs to class 1.

3. Decision Rule

- If $P \geq 0.5$, predict class 1.
- If $P < 0.5$, predict class 0.

The threshold (0.5) can be adjusted depending on the application (e.g., in medical diagnosis, you might prefer a lower threshold to catch more positives).

4. Sigmoid Function Graph

The sigmoid function has an S-shaped curve:

- As $z \rightarrow -\infty$, $\sigma(z) \rightarrow 0$
- As $z \rightarrow +\infty$, $\sigma(z) \rightarrow 1$
- At $z = 0$, $\sigma(z) = 0.5$

This smooth transition makes it ideal for modeling probabilities.

8. How can learning curves help in identifying overfitting and underfitting? Illustrate with examples?

A. **Learning curves** help diagnose overfitting and underfitting by showing how a model's performance changes with increasing training data. They reveal whether the model is generalizing well or failing to learn effectively.

1. What Is a Learning Curve?

A learning curve plots **model performance** (e.g., accuracy or error) on the **training and validation sets** against the **size of the training data** or **number of training iterations**.

- **X-axis:** Number of training examples or epochs

- **Y-axis:** Performance metric (e.g., accuracy, loss)

2. Underfitting

- **Symptoms:**
 - Low training score
 - Low validation score
 - Small gap between training and validation curves
- **Cause:** Model is too simple or not trained enough
- **Example:** Using linear regression on a nonlinear dataset
- **Learning Curve:**
 - Both curves plateau at low performance
 - Increasing data or training time doesn't help

3. Overfitting

- **Symptoms:**
 - High training score
 - Low validation score
 - Large gap between training and validation curves
- **Cause:** Model is too complex or memorizing noise
- **Example:** Deep neural network trained on small dataset
- **Learning Curve:**
 - Training curve remains high
 - Validation curve drops after a point
 - Adding more data or regularization may help

4. Well-Fitted Model

- **Symptoms:**
 - Training and validation scores converge
 - Both curves improve with more data
- **Example:** Random forest on a well-prepared dataset
- **Learning Curve:**
 - Small gap between curves
 - Both curves reach high performance

9. What is Random Forest? Explain its construction using decision trees with an example?

A. 1. Core Concept

- A **decision tree** splits data based on feature values to make predictions.
- A **random forest** builds many such trees using random subsets of data and features, then aggregates their predictions (majority vote for classification, average for regression).

2. Construction Steps

1. Bootstrap Sampling:

- From the original dataset, create multiple random samples (with replacement).
- Each sample is used to train one decision tree.

2. Feature Randomness:

- At each split in a tree, only a random subset of features is considered.
- This reduces correlation between trees and improves generalization.

3. Tree Building:

- Each tree is grown independently and fully (no pruning).
- Trees may overfit individually, but the ensemble smooths out errors.

4. Prediction Aggregation:

- For classification: majority vote across trees.
- For regression: average of all tree outputs.

3. Example

Suppose you're predicting whether a loan applicant will default:

- **Dataset:** Features include income, credit score, loan amount, employment status.
- **Random Forest Construction:**
 - Build 100 decision trees, each trained on a different bootstrap sample.
 - At each split, randomly select 3 out of 10 features.
 - Each tree gives a prediction (default or not).
 - Final prediction is the majority vote across all trees.

```
from sklearn.datasets import load_iris
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report
```

```
# Load dataset
iris = load_iris()
X = iris.data
y = iris.target
```

```
# Split into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
random_state=42)
```

```
# Initialize Random Forest
rf = RandomForestClassifier(n_estimators=100, random_state=42)
```

```
# Train the model
rf.fit(X_train, y_train)
```

```
# Predict on test set
y_pred = rf.predict(X_test)
```

```
# Evaluate performance
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

10. Explain logistic regression. Why is it used for classification instead of linear regression?

A. 1. Definition of Logistic Regression:

Logistic Regression is a supervised machine learning algorithm used for classification problems — especially binary classification (e.g., yes/no, 0/1, spam/not spam). It predicts the probability that an instance belongs to a particular class by using the logistic (sigmoid) function.

2. Logistic (Sigmoid) Function:

The logistic regression model uses the **sigmoid function** to map any real-valued number into a range between **0 and 1**.

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(b_0 + b_1 X)}}$$

Where:

- $P(Y = 1 | X)$: Probability that the output is 1 (positive class)
- b_0 : Intercept term
- b_1 : Coefficient (weight)
- X : Input feature

Sigmoid curve:

- S-shaped curve ranging from 0 to 1
- If $P(Y = 1 | X) > 0.5$, classify as **1** (positive class)
- Otherwise, classify as **0** (negative class)

3. Working of Logistic Regression:

1. Compute the **linear combination**:

$$z = b_0 + b_1 x_1 + b_2 x_2 + \dots + b_n x_n$$

2. Apply the **sigmoid function** to convert z into probability:

$$P = \frac{1}{1 + e^{-z}}$$

3. Set a **decision threshold** (commonly 0.5):

- If $P \geq 0.5 \rightarrow$ Class 1
- If $P < 0.5 \rightarrow$ Class 0

4. Why Logistic Regression Is Used for Classification Instead of Linear Regression

Aspect	Linear Regression	Logistic Regression
Output Range	Can produce any real value ($-\infty$ to $+\infty$)	Always between 0 and 1
Interpretation	Output not meaningful as a probability	Output interpreted as probability
Best for	Predicting continuous values	Predicting categorical classes

Aspect	Linear Regression	Logistic Regression
Problem	Linear regression may predict invalid probabilities (e.g., -0.8 or 1.3)	Logistic regression keeps predictions within 0–1 range

Example:

Suppose we want to predict whether a student passes (1) or fails (0) based on study hours.

- Linear regression may predict values like **1.2** or **-0.4**, which are meaningless probabilities.
- Logistic regression converts predictions into **valid probabilities** (e.g., 0.85 → Pass, 0.15 → Fail).

11. Differentiate between linear and polynomial regression. Why is polynomial regression used for non-linear data?

A.

Aspect	Linear Regression	Polynomial Regression
Model Equation	$y = \beta_0 + \beta_1 x$	$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_n x^n$
Relationship Modeled	Linear relationship between input and output	Non-linear relationship modeled using polynomial terms
Curve Shape	Straight line	Curved line (parabola, cubic, etc.)
Feature Transformation	No transformation; uses raw features	Transforms features into higher-degree terms
Use Case	When data shows a linear trend	When data shows curvature or non-linear patterns
Overfitting Risk	Lower	Higher (especially with high-degree polynomials)
Interpretability	Easy to interpret	Harder to interpret as degree increases
Computational Cost	Low	Higher with increasing degree

Polynomial Regression

- Many real-world datasets show **non-linear relationships**, where data cannot be fitted by a straight line.
- **Polynomial Regression** adds higher-degree terms ($X^2, X^3, \dots$) which allow the model to **capture curves and complex patterns** in data.
- It still uses the **linear regression principle** (linear in coefficients), but models **non-linear relationships** between X and Y.